

Ozmoo

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A Z-machine interpreter for the Commodore 64 and similar computers

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Program and Data Structures

Overview

The main parts of Ozmoos are as follows:

- The interpreter This is the main program that sets up the memory structures, reads the config into memory, and then starts executing the program instruction by instruction in the Z-machine emulator, taking care to read portions of the Z-code file from disk as needed.
- Virtual memory buffers On some computers it is hard to use all the available RAM without time-consuming banking operations. To improve efficiency the virtual memory buffers are used when Ozmoos is compiled with virtual memory support, and located in RAM that is guaranteed to be accessible. By having the buffers in accessible RAM most operations can be done with direct RAM access, and banking is only needed when the program counter leaves the buffer. This system is used on C64 and C128.
- Z-machine stack This is used to store arguments and function calls. The stack is described in detail in a chapter of its own.
- Virtual memory The rest of the available RAM is divided into 512-byte blocks and used to store parts of the story file as needed by the game state. The virtual memory is a complex topic which is covered in more detail in a chapter of its own.

The next figure shows how memory may be allocated when playing a game on the Commodore 64.

Memory map for Ozmoo, playing a Z5 story

2018-10-15

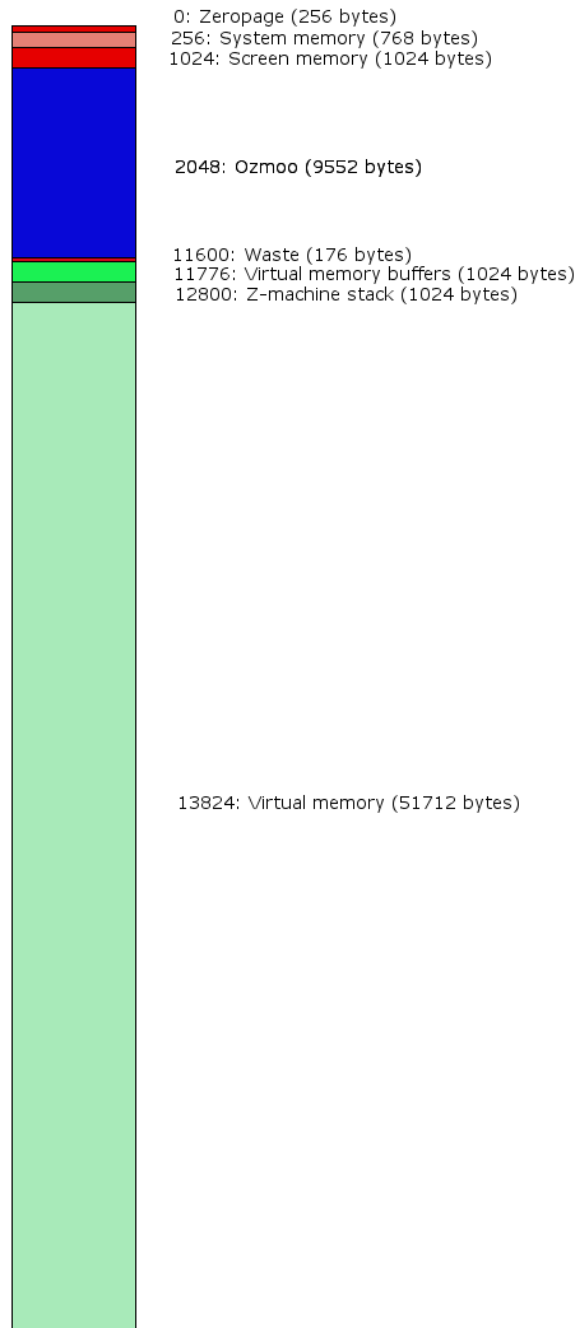


Figure 1: C64 memory overview

Source files

These source files are located in the asm dictory. The table also shows the most important routines included in each source file.

File/Routine	Comment
ozmoo.asm <i>program_start</i>	Ozmoo's main routine where execution of Ozmoo begins
zmachine.asm <i>z_execute</i>	implements the Z-machine execute the Z-machine program
c65toc64wrapper.asm	code to go to C64 mode on the MEGA65
constants.asm	zero page allocations and kernal labels
constants-c128.asm	C128 zero page allocations and kernal labels
constants-header.asm	labels to access the header part of a story file
dictionary.asm	routines to access the dictionary of a story file
disk.asm <i>readblock</i> <i>readblocks</i> <i>read_track_sector</i>	read and write to floppy disk read a memory block from the story file read a memory block from the story file read single sector from the floppy disk
file-name.asm (in temp folder)	holds the name of the boot file (file is generated by make.rb)
file-name.tpl	template used by make.rb to generate file-name.asm
memory.asm	routines to access the Z-machine memory
objecttable.asm	routines to access the object tree defined in a story file
picloader.asm	code to show a picture while reading the story file
reu.asm	implements the Ram Expansion Unit interface for C64 and C128, banked RAM for the X16, and an interface to Attic RAM on MEGA65
screen.asm	printing routines

File/Routine	Comment
screen-z6.asm	printing routines for version 6, with its window model and graphics
screenkernal.asm	low-level display routines
screenkernal-z6.asm	low-level display routines for version 6
sound.asm	routines to support playing sound effects
sound-aiff.asm	routines to support sound effects in AIFF format
sound-wav.asm	routines to support sound effects in WAV format
splashlines.asm (in temp folder)	contains splash screen text (file is generated by make.rb)
splashlines.tpl	template used by make.rb to generate splashlines.asm
splashscreen.asm	show the splash screen
stack.asm	implements the Z-machine stack
streams.asm	implements text streams for the Z-machine
text.asm	routines to read and write text
utilities.asm	various utilities, error handling and debug routines
vdc.asm	low level routines to write text on the C128 in 80 column mode
vera.asm	defines the interface to the VERA chip for text output on the X16
vmem.asm	virtual memory routines, and corresponding routines for non-vmem builds
walkthrough.asm (in temp folder)	contains walkthrough commands for benchmarking (file is generated by make.rb)
walkthrough.tpl	template used by make.rb to generate walkthrough.asm

File/Routine	Comment
zaddress.asm	routines to access data (i.e. not program code) in the Z-machine memory space

Startup

Ozmoo programs are compressed with Exomizer into a file called **story**, which is loaded and run like a Basic program. This file is sometimes referred to as a boot file. When this program is executed, the uncompressed Ozmoo program replaces **story** and execution starts from the `program_start` label.

First the screen is initialised and the splash screen displayed (if any). Then the config blocks (the first two blocks on the boot disk) are read, Ozmoo checks for an REU, the interpreter may ask the player to insert the story disk(s) into the drive(s), Ozmoo may read all of the story file into the REU, or it may load parts of it into RAM (if there is still some empty RAM after uncompressing the boot file). Then the Z-machine program counter is set to the start of the story, as specified in the header, and execution of the Z-machine program begins.

Z-machine

The Z-machine executes instructions, using the Z-machine stack to keep track of calls and arguments. A Z-code file (AKA game file or story file, like `zork1.dat` or `planetfall.z3`) is essentially a snapshot of the memory of a virtual computer (the Z-machine). A Z-code file is divided into dynamic memory (up to 64 KB) and static memory (everything above dynamic memory, up to 512 KB). Dynamic memory is the only part that can be altered during program execution. Ozmoo keeps the entire contents of dynamic memory in RAM at all times, while using whatever RAM is left for virtual memory, to hold as much of static memory as possible. Whenever the player moves into a new location or tries something new and the disk starts spinning, this is because the program needs a part of static memory that isn't in RAM, so it has to be fetched from disk. What's so nice about this system is that it's handled entirely by the interpreter. The author of a game doesn't have to figure out when to load which parts of the story file - they can just write their game as if it's going to run on a computer with 128 or 256 KB of RAM, and the interpreter makes it work on a computer with much less RAM than that.

To further conserve space, the Z-machine has a text compression scheme built in. Basically it means each character is translated into one or more five-bit codes. By default, the space character and all lowercase letters fit into a single

five-bit code each, while uppercase letters and special characters require two or four five-bit codes. These five bit codes are then grouped together so three of them occupy two bytes. I.e. “adventure” is stored as “adv” in two bytes, “ent” in two bytes and “ure” in two bytes, thus using six bytes. On top of this, there is a system of abbreviations, so common strings can be replaced by two five-bit codes.

Disk I/O

Ozmoo is designed to use a floppy disk layout similar to what Infocom used, but it has some extra options such as using multiple disk drives to play bigger games. Config information for the game is stored in sector 0 and 1 of track 1 on the boot disk. The static memory portion of the story file is stored in the track 1 and up on the story disk(s). The boot file, containing the interpreter, dynamic memory and as much of static memory as possible, is stored on the boot disk or in the remaining free space of the combined boot / story disk.

The disk I/O routines are located in disk.asm. The virtual memory uses read_block and read_blocks to read data from the story file when needed. There is a routine read_track_sector for reading a single sector of the story file specified by track number and sector number.

REU

If Ozmoo detects a RAM Expansion Unit (REU), and it’s big enough to hold the entire story file, the player gets the question if they want to use it. If they do, the static part of the story file is read into the REU before the game starts. When the virtual memory system needs to read data from the story file by calling read_block or read_blocks (disk.asm) it will read the blocks from the REU instead of the floppy drive.

The interpreter detects the size of the REU in banks, where a bank is 64 KB. After deciding whether to use the REU for caching the story file, it may also allocate a bank for undo and a bank for scrollbar buffer, if there is room and the game was built with support for these features. If the story file is stored in REU, it always begins in bank 0 and may use up to 8 banks. If story data is stored in the REU, the first page of bank 0 is reserved for fast page copying operations. If story data is not stored in the REU, and the REU is 256 KB or bigger, the first page of bank 3 is used for fast page copying operations instead. If the REU is smaller, fast page copying isn’t used.

X16 builds

On the X16, the main game file doesn't hold any story data. The entire Z-code file is held in a file called "zcode" and is loaded into banked RAM on boot. Since the entire file is held in one linear chunk of memory, and accessed in place (using banking), this is not a virtual memory build, and the VMEM constant is not defined. No config information is needed, so there are no config blocks on disk. This means a X16 game can be copied from one disk to another using a regular file copier.

X16 Memory map

Similar to C128, the graphics output is handled by a chip called VERA that contains the video memory. This means that no normal memory is needed for screen RAM. There is a standard 512 KB extended memory, of which a chunk/bank at a time is available at \$a000-\$bfff. The currently accessible bank of the extended memory is controlled by setting \$0000 to the chunk number.

Normal memory

Address range	KB	Usage
\$0000-\$9fff	40	System RAM, interpreter
\$a000-\$bfff	8	Story file (currently banked chunk)

Video memory (VERA)

The standard text mode needs $80 \times 60 \times 2 = 9600$ bytes, starting from \$1b000, since the screen is 80 columns, 60 rows, and each character is stored as a (character code, colour) tuple.

Address range	KB	Usage
\$1b000-\$1d580	about 10	Screen RAM

Extended/banked memory

Address range	KB	Usage
\$00000-\$02000	8	Reserved space
\$02000-\$7fff	504	Story file

The extended memory is mapped into \$a000-\$bfff, so the memory is accessible in 8KB chunks. Note that the first 8KB is used by the kernal and DOS routines, so we load the story file from \$02000.

MEGA65 builds

On the MEGA65, the boot file doesn't hold any story data. The entire Z-code file is held in a SEQ file called "zcode" and is loaded into Attic RAM on boot. Since the entire file is held in one linear chunk of memory, and accessed in place, this is not a virtual memory build, and the VMEM constant is not defined. No config information is needed, so there are no config blocks on disk. This means a MEGA65 game can be copied from one disk to another using a regular file copier.

A version 6 game built with Z6_PICTURES also has one file per picture on the disk, named p001 to p255, which are decompressed into Attic RAM at boot. See "Version 6".

MEGA65 Memory map

Address range	KB	Usage
\$0000-\$ffff	64	System RAM, screen RAM, interpreter
\$10000-\$1ffff	64	Picture tile store (version 6, Z6_FCM_MODE)
\$40000-\$40fff	4	Screen RAM for scrollback mode
\$40000-\$4ffff	64	Current sound effect
\$50000-\$5ffff	64	Undo buffer
\$08000000-\$0807ffff	512	Story file
\$08080000-\$080800ff	0.25	Signature of last game that was loaded (to make restart faster)
\$08100000-\$081ffff	1024	Sound data
\$08200000-\$082ffff	1024	Scrollback buffer
\$08300000-		Pictures, decompressed (version 6, Z6_PICTURES)
\$0ff80000-\$0ff807ff	2	Colour RAM
\$0ff80800-\$0ff817ff	4	Colour RAM for scrollback mode

Bank 1, \$10000-\$1ffff, is untouched in an ordinary build: nothing in Ozmo addresses it, and the C64 mode kernal leaves it alone. A version 6 build in Full Colour Mode uses it as the tile store, which holds 1024 tiles of 64 bytes. It is the only part of fast RAM the VIC-IV can be pointed at that is not already spoken for: \$20000-\$3ffff is the ROM, \$40000-\$4ffff the current sound effect and

\$50000-\$5ffff the undo buffer. A picture may therefore have at most 1024 unique tiles, which is enough for a full 40x25 screen in which no two cells are alike.

In Full Colour Mode a cell is two bytes in both screen RAM and colour RAM, so the 40x25 screen uses the same 2000 bytes as the 80x25 text screen, and the map above does not change.

Note that the scrollbar buffer's screen RAM and the current sound effect both begin at \$40000. Opening the scrollbar while a sound is playing therefore corrupts the first 4000 bytes of the sample. It repairs itself, because the next @sound_effect copies the sample from Attic RAM again.

Save and Restore

The story file is read piece by piece by mapping the program counter to the correct track and sector, and read it into memory. The main function doing this is readblocks located in disk.asm

disk.asm also contains save and restore functionality. The main functions are do_save and do_restore. The save files are normal files that contain some important internal variables such as the program counter, the Z-machine stack, and the dynmem part of the RAM.

Undo

If extra memory is available then it can be used to support undo for games that probe for this functionality. This is currently supported for C64 and C128 with a RAM Expansion Unit (REU), and for the MEGA65. To enable undo support the `make.rb` script should be called with the `-u` switch. If undo is enabled and the header has the undo flag set, then Ozmoos checks for available memory. If not found, then the undo header flag is cleared, and an error message is shown.

When undo support is active then the current game state (the stack, dynamic memory and some variables) are copied into the undo buffer before a new command is read from the user. This is similar to the save command, but saves to memory instead of saving to a file. If the user gives the undo command, then the state (dynamic memory, stack etc.) is instead copied from the undo buffer, and the execution continues from the saved state, undoing the previous turn.

The main functions for undo are `do_save_undo` and `do_restore_undo` in `disk.asm`. As in Frotz, a hotkey (Ctrl-U) has been added to enable undo support for z3 games, even if official undo support is only available for z5 and upwards. The hotkey handling code is found in `text.asm`.

Screenkernel

Screenkernel, implemented in `screenkernel.asm`, is a replacement for the low-level screen output routines on the Commodore computers. It is needed to abstract away low-level implementation details so that new targets can be more easily added, and also to support custom text scrolling to efficiently enable status lines, especially big ones used in games such as *Border Zone* and *Nord and Bert*. The number of lines to protect when scrolling is stored in `window_start_row`

The main functions of screenkernel are `s_printchar`, which replaced `CHROUT $FFD2`, and `s_plot` which replaced `PLOT $FFF0`. It also provides `s_set_textcolour`, `s_delete_cursor`, `s_erase_window` `s_erase_line`, `s_erase_line_from_cursor` and `s_init`.

Screenkernel is started by calling `s_init`. Internally it keeps track of the cursor position so it can put a character on the screen when `s_printchar` is called.

For the Commodore 64 version the characters are stored directly in the video memory, and the colour in the colour memory. The Commodore 128 version detects if 40 or 80 columns mode is used while running the program. If 40 characters are used, then it works like the Commodore 64 version. But if 80 columns are used, then the C128's Video Display Controller chip (VDC) is used. Instead of writing directly into the video memory, character output, scrolling and other screen commands are sent by VDC registers. The file `vdc.asm` contains functions that make this communication easier.

The X16 is using a separate chip called VERA for text output, similar to VDC for the C128. The file `vera.asm` defines the interface to the VERA chip used for putting characters on the screen.

The Plus/4 and Commodore 128 in 80 column mode doesn't use the same palette as the Commodore 64. Mapping tables (`plus4_vic_colours` and `vdc_vic_colours`) are used to assign C64 colours to their closest equivalents on these platforms.

Version 6

Version 6 of the Z-machine replaces the two windows of versions 3 to 5 with eight, each with its own rectangle on the screen, its own cursor and its own colours, and it expects the interpreter to draw pictures. Ozmoos implements the window model on every target and the pictures on the MEGA65.

Because so much of the screen layer changes, version 6 gets its own copies of the two screen files: `screen-z6.asm` and `screenkernel-z6.asm`, sourced instead of `screen.asm` and `screenkernel.asm` when the Z6 flag is set. They are meant to stay close to the files they were forked from; when the originals change, it is better to fork them again than to patch the copies by hand.

The window model

The eight windows are held in the property arrays the standard describes (`window_y`, `window_x`, `window_y_size`, `window_x_size`, and so on), laid out one after another so that `get_wind_prop` and `put_wind_prop` can find a property with `window_y + 8 * property + window`. Coordinates are stored 0-based; the opcodes convert to and from the 1-based coordinates the Z-machine uses.

Property 13, the font size, is the only property that is a real 16-bit word: the height in the high byte and the width in the low one. It cannot live in the byte-per-window arrays with the others, so `get_wind_prop` answers it directly, with 1 and 1, because the header tells the game that a character is one unit wide and one unit high. A game may divide by it, so it must never be zero.

Printing, wrapping, scrolling, the cursor, the MORE prompt and erasing are all per window: text wraps at the window's right margin, a window scrolls its own rectangle, and the MORE prompt appears in the window's bottom right cell rather than the screen's. Each window keeps its own cursor and its own line count.

The window-local scrolling only covers the code path the C64, Plus/4 and MEGA65 share (`.s_scroll`, and `s_scroll_window` for the `scroll_window` opcode). The C128's 80 column routine and the X16's still scroll the whole screen, so a window on those targets will smear the rest of the display.

Extended Color Mode on the C64

The C64 has one background colour, which is no use to a game that wants eight windows in eight colours. `Z6_ECM_MODE`, selected with `make.rb`'s `-ecm` switch, turns on the VIC-II's Extended Color Mode: the top two bits of a screen code choose one of four background registers, `$d021` to `$d024`, so each window can have its own background. The price is that only 64 characters remain, so screen codes are masked to six bits, uppercase renders as lowercase, and reverse video is not available. ECM is C64 only; `make.rb` rejects it on other targets.

Full Colour Mode on the MEGA65

The MEGA65 does not need the ECM compromise. `Z6_FCM_MODE`, selected with `-fcm`, puts the VIC-IV into Full Colour Mode with 16-bit character codes: a 320x200 screen of 40x25 cells, which is the resolution the Amiga versions of the version 6 games were drawn for.

The registers are set up in `init_mega65`, with the hot registers turned off so the values stick. The important ones are `$d031`, where `H640` and `V400` are cleared and, less obviously, `ATTR` must also be cleared, because that is what selects 8-bit colour; and `$d054`, where `CHR16` and `FCLRHI` are set while `FCLRLO` is left clear. `FCLRLO` clear is the whole trick: screen codes below 256 are still ordinary 8-byte glyphs fetched from `CHARPTR`, so all the text renders as before, while codes from 256 up are 64-byte full colour tiles. `CHARPTR` points at `$2d000`, the C64 font in the MEGA65's ROM, whose codes 128 to 255 are the reversed glyphs, so reverse video keeps working.

A cell is two bytes in both screen RAM and colour RAM, so a row of 40 cells is the same 80 bytes as the 80 column text screen it replaces, and neither buffer grows. `Ozmoo` writes the character into the even byte of a screen cell and the colour into the odd byte of a colour cell. The other two bytes, the screen cell's high byte and the colour cell's flags, must be zero and stay zero: `init_mega65` clears them once.

The code that reaches every cell would otherwise have to change everywhere, so:

- `zp_screencolumn` still holds a column, and only the sites that index the screen double it. That leaves the window edges, the margins and every comparison alone.
- `zp_colourline` is biased by one, so the same doubled index writes both the character and the colour. A row starts at a multiple of 80, so the bias never carries.

- Every place that writes a character also writes zero to the cell's high byte, with the `clear_cell_high_byte` macro. Without it, a character printed over a picture would leave the cell pointing at a tile.

Note that most text does not pass through `s_printchar` at all: it goes through `print_line_from_buffer` in `screen-z6.asm`, which writes the screen directly.

Pictures

Pictures are drawn only on the MEGA65, and only with `Z6_FCM_MODE`. On the C64 and the Plus/4, `draw_picture` writes a “pic:N” note where the picture belongs, straight to the screen so that it never reaches the transcript, and `erase_picture` paints the note out.

The pictures are converted from PNG by `tools/pics2asm.py`, which writes one `p<nnn>.bin` per picture. `make.rb`'s `-pics` switch runs it, puts the files on the d81 next to `zcode`, and sets `Z6_PICTURES`. Only an index of picture numbers is assembled into the interpreter, so a set of a hundred and forty pictures costs it nothing. At boot, `pic_load_all` reads each file and decompresses it into Attic RAM, in the same spirit as `read_sound_files`, though it opens and reads the files itself rather than calling `m65_load_file_to_reu`, because it has to decompress on the way.

A picture file holds the cell dimensions, the number of unique tiles, a 48-byte palette, a cell map of two bytes a cell, and then the tiles. Identical cells are stored once. It is RLE compressed, PackBits style: a token of 0 to 127 is followed by that many plus one literal bytes, and a token of 129 to 255 is followed by one byte repeated 257 minus token times. Decoding needs no lookahead.

Two things about the pixels:

- A tile is 8x8 pixels and a pixel is four bits, so two pixels share a byte on disk and in Attic RAM. In the tile store a pixel is a whole byte. Storing four bits a pixel halves the picture set before it is compressed, and Arthur's does not otherwise fit on a d81.
- A colour index of 0 is transparent, and 255 is taken from colour RAM, so neither may be a real colour. A picture therefore has at most 15 colours.

Drawing a picture reads its header and palette straight out of Attic RAM, gives the window a run of the tile store, copies the tiles down into it, and fills the window's cells. In Full Colour Mode a cell's screen code is its tile's address divided by 64, and the tile store begins at \$10000, so a code is simply \$0400 plus the tile's index.

The palette bank belongs to the window, not to the picture. A window holds at most one picture, so the eight banks of 16 entries above the 16 text colours are

enough for any number of pictures, and a window's bank is $16 + 16 * \text{window}$. A picture's pixel indices are stored relative to entry 16, so the tile copy adds the window's offset to each pixel as it goes, leaving a pixel of 0 alone because it is transparent in every bank. The DMA engine can neither expand a nybble nor add, so that copy is done by the CPU; at 40 MHz even a full screen picture is a blink.

Each window keeps the run of the tile store it was given, and reuses it when the next picture fits. When the store runs out the allocator starts again at the bottom, which can only spoil a picture in another window.

The stack

The exact layout of the stack is an implementation detail, not part of the Z-machine specification. However, the Z-machine specification does say a few things about how the stack should *work*. Each routine call creates a stack frame. Instructions can push any number of values onto the stack, and pull values from the stack, or read or change the top value without pushing or pulling it. A value is always word-sized. Pushed values can only be accessed by instructions in the same routine (= same stack frame). When a routine returns, all values which have been pushed inside that routine and are still on the stack are discarded. The stack resides outside the Z-machine, and there is no way for a Z-machine program to check if there are values on the stack, if there is room for more values, how many stack frames there are etc.

The stack in Ozmoos consists of a number of words. The default size is 512 words. With each procedure call that is made, a new stack frame is created. The first stack frame begins at the first address of the stack (label `stack_start`).

A stack frame consists of the following parts:

1. The current values of all local variables of the routine (0-15 words).
2. Frame header (2 words)

Frame header A has the following parts:

Byte 0, bit 7: 1 = The return value of this procedure call should be written into a variable

Byte 0, bit 4-6: In version 5+, the number of parameters this procedure was called with (0-7)

Byte 0: bit 0-3: The number of local variables this procedure has (0-15)

Byte 1, bit 7: 1 = this routine call was done due to an input interrupt

Byte 1, bit 0-2: Bit 16-18 of the return address

Frame header B:

Bit 0-15 of the return address (highbyte first).

3. Pushed words (0 or more words)
4. Number of pushed bytes (1 word)

The first frame lacks part 1 and 2. The current frame lacks part 4.

Apart from this, Ozmoos also has a few variables on zeropage to keep track of the stack:

`stack_ptr` (word): a pointer to where part 3 of the current stack frame begins

`stack_pushed_bytes` (word): A counter of how many words part 3 of the current frame consists of.

`stack_has_top_value` (byte): If the value is 1, the last value pushed onto the stack in the current frame hasn't been put on the actual stack, but resides in the variable `stack_top_value` instead.

`stack_top_value` (word): If `stack_has_top_value` is 1, this contains the top value on the stack.

The `stack_top_value` variable has been added for performance reasons.

Frame 0 isn't a full stack frame. When a Z-machine interpreter begins execution of a program, it just starts at the start address given in the header of the story file, with an empty stack. The start address can be in the middle of a routine. This means the interpreter doesn't know of any local variables, and there is no return address - it's not legal to perform a return from this routine. This "fake" routine is just a simple way to get started, and the only thing you want to do in this routine is call a "real" routine and then perform quit. Inform 6 adds such a routine automatically, and it calls the routine "main" which must be defined in the Inform 6 program and then quits. To make things more confusing, this little "fake" routine is called "main" when disassembling the Z-code program.

Let's go through an example program and what it does to the stack. We run it until it performs `@read_char` and then we pause and look at the stack. This is a version 5 program.

Inform's built-in "fake" main routine (on address \$4ec) calls our main routine (on address \$514). This causes Omzoo to close the current frame (Frame 0, which isn't a full stack frame) by moving the top value from `stack_top_value` (if any) to the stack, and writing the total number of pushed bytes as a word to the stack. In this case, there are no values on the stack, and no value in

```

!% ~~$
!% $OMIT_UNUSED_ROUTINES=1

[foo x y;
  @push 9;
  @push 3;
  print x;
  new_line;
  print "Press a key: ";
  @read_char y;
  print y;
];

[main;
  @push 7;
  foo(5);
];

```

Figure 2: stackdemo.inf, Inform 6 source code

```

Main routine 4ec, 0 locals

4ed: e0 3f 01 45 ff      CALL_VS      514 -> Gef
4f2: ba                  QUIT

Routine 4f4, 2 locals

4f5: e8 7f 09            PUSH        #09
4f8: e8 7f 03            PUSH        #03
4fb: e6 bf 01            PRINT_NUM   L00
4fe: bb                  NEW_LINE
4ff: b2 ...             PRINT        "Press a key: "
50a: f6 ff 02            READ_CHAR   -> L01
50d: e6 bf 02            PRINT_NUM   L01
510: b0                  RTRUE

Routine 514, 0 locals

515: e8 7f 07            PUSH        #07
518: da 1f 01 3d 05      CALL_2N     4f4 (#05)
51d: b0                  RTRUE

```

Figure 3: stackdemo.zasm, disassembly

Stack contents:			
3a00:	00 00	Number of bytes pushed	Frame 0
3a02:	80 00	Frame header A	
3a04:	04 f1	Frame header B	
3a06:	00 07	Pushed word 0	Frame 1
3a08:	00 02	Number of pushed bytes	
3a0a:	00 05	Local variable 0	
3a0c:	00 00	Local variable 1	
3a0e:	12 00	Frame header A	Frame 2
3a10:	05 1d	Frame header B	
3a12:	00 09	Pushed word 0	
stack_ptr: 12 3a			
stack_pushed_bytes: 00 02			
stack_has_top_value: 01			
stack_top_value: 00 03			

Figure 4: stackdemo, stack contents

stack_top_value, so Ozmoos just copies the value in stack_pushed_bytes (0) to the stack.

Ozmoo then starts a new stack frame - frame 1. The main routine at \$514 has 0 local variables, so it doesn't reserve any space for part 1. It then writes the frame header to the stack. Frame header A is \$80, \$00 because the procedure call is made using call_vs which expects the return value to be written into a variable, there are no arguments and the routine has no local variables, the routine was not called due to an interrupt, and bit 16-18 in the return address are all 0. Frame header B holds the value \$04f1, which points to the last byte of the call_vs statement, holding the variable number where the result is to be stored upon return.

Program execution now continues in our main routine at address \$515 (the routine's official start address is 514, but the first instruction is on address \$515.) The first instruction pushes the value 7 onto the stack. The second instructions performs a call_2n instruction to call the foo routine with one argument, which is the value 5. We can see that the value 7 has been pushed onto the stack, and that the counter says a total of 2 bytes have been pushed onto the stack in this frame.

Ozmoo now opens stack frame 2. The routine being called (the foo routine located at address \$4f4) has two local variables, so Ozmoos makes room for two words on the stack for part 1 of the frame. The first variable gets the value 5, since the procedure was called with this as an argument. Ozmoos then adds part 2, the header. Frame header A gets the value \$12, \$00 because the routine was not called with an instruction which expects the return value to be written into a variable, the routine call was made with 1 parameter, the routine has 2 local variables, the routine was not called due to an interrupt, and bit 16-18 of the

return address are 0. Frame header B holds the value \$051d, which is where execution should resume upon return from this routine.

The foo routine pushes the value 9 and then the value 3. At the time when we inspect the stack, the value 3 is held in `stack_top_value`. If we were to push another value, or call a procedure, this value would be written to the stack as well.

There are two more `print` instructions and a `new_line` instruction before `read_char`, but none of them interfere with the stack. When we hit `read_char`, we paused the program and inspected the stack.

Virtual Memory

This chapter is based on a document written by Steve Flintham.

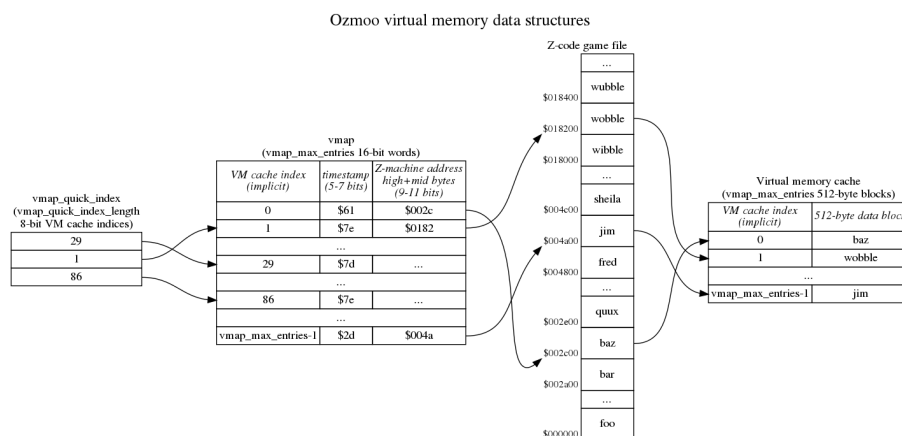


Figure 5: Virtual memory overview

The basics

The virtual memory subsystem is only used for the game's static and high memory, which are read-only. The game's dynamic memory is always held entirely in RAM.

The virtual memory code does most of its work in the `read_byte_at_z_address` subroutine. (This can be seen in `vmem.asm`; there are multiple versions of this subroutine in the file because conditional assembly is used to allow non-VM builds, but the VM version starts about halfway down.) It's entered with a 24-bit Z-machine address in A, X and Y (high byte in A, middle byte in X, low byte in Y) and returns the value at this address in A. Also, upon returning, Y is 0 and `mempointer` (a 2-byte zero page pointer) is set so that `"lda (mempointer),y"` returns the byte at the 24-bit address given.

Note that `read_byte_at_z_address` isn't called for every single read from Z-machine high and static workspace; there are subroutines layered on top of it which only call it when necessary.

Virtual memory is handled in 512-byte blocks, which are always aligned at a 512-byte boundary in the game file and in memory. This means that every VM block has a Z-machine address of the form `$abcd00`, and the least significant bit of `d` is always 0.

The VM system has a cache with `vmap_max_entries` 512-byte blocks of RAM to use to hold blocks read from disc. There's a parallel data structure called the `vmap` with `vmap_max_entries` 16-bit words in it to track what block of game data is in each cache block. At the most basic level, we can imagine that if block 4 of the cache contains the 512-byte block starting at `$018200` in the game file, `vmap[4]` contains `$0182`.

If we want to access the byte at `$018278`, we round that down to the previous 512-byte boundary to get `$018200` and then search through all the entries in `vmap` to see if any of them contain `$0182`. In this case, entry 4 does, so we'll set `mempointer` to `cache_start_address+4*512+$078`, load the value at that address into `A` and return. Where does the `$078` come from? This is the low 9 bits of the address we were given, which can be thought of as an offset to the byte of interest within this 512-byte block.

What if none of the `vmap` entries contains `$0182`? In that case we need to pick one to overwrite; let's say we pick entry 7. Because we only use virtual memory for read-only data, we can just read the 512-byte block at offset `$018200` in the game file into memory at `cache_start_address+7*512` (overwriting whatever was there), set `vmap[7]` to `$0182` and return with `A` set to the value held in address `cache_start_address+7*512+$078`.

At the most basic level, that's all there is to it. But in practice there are some additional details we need to take care of.

Timestamps

We need to be able to make a sensible decision about which cache block we're going to overwrite when we need to read a block of data in from disc because it's not in the cache. What we want to do is keep hold of cache blocks we've recently used, and instead discard a block we haven't used in a while, on the reasonable assumption that we're more likely to use a block in the near future if we've recently used it.

To implement this, each `vmap` entry also contains a timestamp. There's a global "current time", called `vmem_tick`, and every time we access a cached block its timestamp is set to `vmem_tick`. `vmem_tick` is incremented whenever we need to read data from disc because it's not in the cache. Note that several entries

in the vmap can therefore share the same timestamp - if we access cache blocks 4, 8 and 22 without needing to read anything from disc, all of those will have the same timestamp (the current value of `vmem_tick`).

Also, there's a mechanism to keep `vmem_ticks` from getting too big for the space available. For a z3 game, we have 8 bits for storing tick values for each vmem block. We start with the value 0 and we can just increase it until we reach 255. When we hit 256, we change the tick counter to 128, and we reduce the tick value in all vmem blocks by 128. Since we're using unsigned numbers here, a block which had the tick value 27 now gets 0. A block which had tick value 150 now gets 22 etc. The next block we read from disk gets the tick value 128, the next after that gets 129 etc. Next time we reach 256, we just repeat the above procedure.

With that infrastructure in place, when we need to overwrite a block in the cache with a new block from disc, we can pick a block with the oldest timestamp. (Not *the* block with the oldest timestamp, because as noted above several blocks can share the same timestamp.) There's one caveat, which is that if the Z-machine program counter is currently pointing into a cache block, it is exempt from being overwritten - it would obviously be a bad thing to overwrite the instructions currently being executed with some arbitrary data! I won't go into too much detail on this here, because it's probably best discussed in the context of the routines layered on top of `read_byte_at_z_address`.

The timestamps are packed into the high bits of the 16-bit entries in vmap, with the low bits representing the high and mid bytes of the Z-machine address. For Z3 games, where Z-machine addresses only have 17 bits (a maximum game size of 128K), only 9 bits of the vmap entry are needed for the high and mid bytes of the address and 7 bits are available for the timestamp. But note that the lowest bit of the address is always zero since the memory is divided into 512-byte blocks. Assuming this zero we can instead store the address in 8 bits and increase the time stamp resolution to 8 bits.

Larger versions of the Z-machine need more bits for the high and mid bytes so fewer bits are available for the timestamp; a Z8 game (with 19 bit addresses for a maximum game size of 512K) only has 6 bits available for the timestamp. This limited timestamp resolution is the reason why `vmem_tick` is only incremented when a block needs to be read from disc, not every time `read_byte_from_z_address` is called.

`vmem_tick` is actually held in memory "pre-shifted" for convenience of using it to compare or update the high byte of the 16-bit entries in vmap, so rather than always incrementing by 1 at a time, it increments by 1 at a time for Z3 games, 2 at a time for Z4 and Z5 games and 4 at a time Z8 games. (In the code, the increment is a constant called `vmem_tick_increment`.)

The quick index

In general we have to do a linear search of vmap in order to see if a particular 512-byte block of the game is already in RAM (and where in RAM it is, if it's in RAM). The vmap can contain up to 255 entries (depending on platform), and it's likely to contain at least 64 entries, representing 32K of cache, so this is potentially quite slow.

It's quite likely that there's a relatively small “working set” of 512-byte blocks which we're going to be accessing over and over again. For example, maybe one 512-byte block contains a Z-machine function with a loop in, and inside that loop we call another Z-machine function which lives in a separate 512-byte block.

We therefore maintain a “quick index” containing the cache indices of the blocks we've accessed most recently. (There are `vmap_quick_index_length` entries in this list, which in practice means there are 6 entries.) We look at the corresponding entries in vmap first to see if those entries have the 512-byte block we're interested in, and if they do we can avoid doing the full linear search. Whenever we have to do the full vmap search, we overwrite the oldest entry in the quick index with the index we found from the full search. The quick index is effectively a circular buffer of `vmap_quick_index_length` entries, with `vmap_next_quick_index` pointing to the oldest entry.

As a consequence of this, the vmap entries pointed to by the quick index are those with the most recent timestamps.

Efficient access to vmap

Each vmap entry is a 16-bit word, so the obvious way to store vmap would be:

```
low byte of entry 0
high byte of entry 0
low byte of entry 1
high byte of entry 1
...
```

The disadvantage of this is that when we're using index registers to step through vmap, we need to do double increment (or decrement) operations:

```
ldy #0
.loop
lda vmap,y ; get low byte of entry
lda vmap+1,y ; get high byte of entry
...
```

```

iny
iny
cpy #max_entries*2
bne loop

```

We don't actually need the two bytes to be adjacent in memory, and so instead we have two separate tables. `vmap_z_l` stores the low bytes:

```

low byte of entry 0
low byte of entry 1
...

```

and `vmap_z_h` stores the high bytes in the same way.

With this arrangement, we don't need double increments (or decrements) to step through the table:

```

ldy #0
.loop
lda vmap_z_l,y ; get low byte of entry
lda vmap_z_h,y ; get high byte of entry
...
iny
cpy #max_entries
bne loop

```

This also has the advantage that we can access up to 256 entries using our 8-bit index registers, instead of 128 entries if we had to do double increment/decrement. The Commodore 64 version of Ozmoos doesn't need this, but the Commodore 128 version does. Acorn sideways RAM version benefits from it and there are a couple of minor tweaks to the code to make this work. (If you know there are <128 entries, you can use `dex:bpl` to control looping over the table and avoid needing a `cpx #255` to detect wrapping.)

Banking

On the Commodore 64 and Commodore 128 there's an additional wrinkle because some blocks of RAM are hidden behind the I/O area and the kernal ROM and there's a mechanism to copy those blocks of RAM into the so-called vmem cache in always-visible RAM when necessary.

The `first_banked_memory_page` variable stores the high byte of the first 512 block of RAM that isn't always visible. On the C64 this is `$d0`, since the

I/O registers are located from \$d000, and unrestricted reading/writing to these memory position cause all kinds of trouble.

When `read_byte_at_z` address detects that we want to read data from a page in the non-accessible memory (\$d000-\$ffff) then we will copy that page (256 bytes) from the non-accessible memory to one of the pages in the vmem buffer. This is done by `copy_page` (in `memory.asm`), which can copy a page securely from and to any memory position using memory banking as needed.

Continuous virtual memory optimization

Copying pages to the vmem buffer is rather expensive, so if possible, we would like the pages that are used the most to be in regular, unbanked RAM, where no such copying is necessary to access the data.

Ozmoo has a feature called continuous virtual memory optimization, which can help with this. It keeps statistics on how many times each block in virtual memory has been requested. After 256 block requests have been made, it swaps the most used block in banked RAM with the least used block in unbanked RAM. Then it resets the statistics, and repeats the cycle. After about 20 such swaps (45 on C128), it slows down the pace and starts waiting for 768 block requests before making a swap.

This ensures that the central parts of the game are gradually moved to unbanked RAM within the first 10-20 moves or so. After that, the system will still be able to adapt to other parts of the game file becoming more important, but it happens at a slower pace.

REU Boost

When Ozmoo is running on a Commodore 64 or 128 with an REU to cache game data, the virtual memory model is no longer optimal for the task - copying data from the REU is so fast, that it's not worth the trouble of tracking when each block was last accessed etc.

With the REU Boost feature, we include a simpler virtual memory model. Essentially, it's a circular buffer, holding a few KB of data. If a page we need can't be found in the buffer, we replace the page that has been in the buffer the longest with the page we need.

Ozmoo decides when the game starts, if game data will be cached to an REU. If so, and the interpreter has been built with the REU Boost feature, it switches to the REU Boost virtual memory model.

Sound

While several Infocom games would play high and low-pitched beeps, a few games had extended sound support using sample playback. Ozmoo supports the basic sound effects (beeps) on all platforms, and extended sounds on the MEGA65.

Sound support is implemented in `sound.asm`. Extended sounds also use `sound-wav.asm` to parse sample files stored in the WAV format. The WAV files need to be 8 bit, mono. Audacity can be used to export wav files in the correct format:

- Select “File/Export/Export as WAV” from the main menu
- Select “Other compressed files” as the file type
- Select “Unsigned 8-bit PCM” as the encoding
- Save the file

If extended sound is to be used, then `make.rb` must be called with the `-asw path` switch. If set, then all `.wav` files in the `path` will be added to the `.d81` floppy created for the MEGA65, and the `SOUND` assembly flag will be set when building Ozmoo.

For legacy reason AIFF is also supported, using `sound-aiff.asm`, and the `-asa path` switch. Code has also been added to detect and handle sound effects in *The Lurking Horror* game from Infocom, which isn’t standard compliant.

When compiled with extended sound support, Ozmoo will preload all sound files during startup into the MEGA65’s attic memory, and then copy each sound effect into fast memory on demand when the `@sound_effect` command is used in the game code.

Smooth Scrolling

This feature adds smooth scrolling support for the c64 target. When active, text is scrolled up one pixel (raster line) per frame rather than an entire character (text row) at a time, providing a “smooth” visual experience. The user can toggle whether smooth scrolling is active using the F2 key during the game.

Smooth scrolling is activated at program startup if the support was included. The code for the feature is in the new file `smoothscroll.asm`. Moving the screen data is performed using the existing scrolling code in `screenkernal.asm`, while the fine-scrolling manipulation is performed in a raster interrupt. This means that some program processing, including printing to the screen, could occur while scrolling is being performed. This helps keep an overall smooth effect when multiple lines are being printed and scrolled (such as with a long room description), but it also means that some coordination is necessary to prevent visual glitches when:

- the program wants to scroll again while scrolling is in progress
- the program will disable interrupts
- the program will perform REU access, which pauses the CPU

One mechanism provided to handle this is the `wait_smoothscroll` function, which can be called to wait until any current scrolling activity has completed. Another is the `smoothscroll_off` and `smoothscroll_on` functions, which can be used to temporarily disable and later re-enable smooth scrolling. The latter are used in the macros `disable_interrupts` and `enable_interrupts`, which are typically used in place of explicit `sei` and `cli` instructions.

The inclusion of smooth scrolling adds about 460 bytes to the interpreter size.

Accented Characters

Ozmoo has some support for using accented characters in games. Since the Commodore 64 doesn't really support accented characters, some tricks are needed to make this work.

The Z-machine, which we are emulating, uses ZSCII (an extended version of ASCII) to encode characters.

The Commodore 64 uses PETSCII (a modified version of ASCII) to encode characters. PETSCII has 256 different codes (actually less, since some code ranges are just ghosts, i.e. copies of other code ranges). Some PETSCII codes are control codes, like the ones to change the text colour or clear the screen. 128 of the PETSCII codes (plus some ghost codes) map to printable characters. A character set contains these, plus reverse-video versions of all 128, adding up to 256 characters all in all.

So, all in all there are basically 128 different characters, and that's it. They include letters A-Z, digits, special characters like !#\$%& etc, some graphic characters, and then either lowercase a-z or 26 more graphic characters. There are no accented characters.

To build a game with Ozmoo with accented characters, we typically need to:

- Decide which (less important) characters in PETSCII and in the C64 character set will be replaced with the accented characters we need.
- Create a font (character set) where these replacements have been made
- Create mappings from PETSCII to ZSCII for when the player enters text
- Create mappings from ZSCII to PETSCII for when the game outputs text

There are already fonts in place for some languages (see documentation/fonts.txt). You can also supply your own font.

The character mappings are created at the beginning of the file streams.asm. There are already mappings for the languages which there are fonts for (see the chapter on fonts in the Ozmoo manual).

To build a game using accented characters, the make command may look like this:

```
ruby make.rb examples\Aventyr.z5 -f fonts\sv\PXLfont-rf-SV.fnt -cm:sv
```

where

-f sets the font to use.

-cm sets the character mapping to use.

If you want to create a character mapping for say Czech, and you know you will never want to build a game in Swedish, you can just replace the Swedish mapping in streams.asm and use -cm:sv to refer to your Czech mapping.

The definition of ZSCII can be found at: <https://www.inform-fiction.org/zmachine/standards/z1point1/sect03.html#eight>

The accented characters which are available by default, and which Ozmoos can use, are in a table under 3.8.7.

The definition of PETSCII can be found at: <http://sta.c64.org/cbm64petkey.html>

Please read the notes below the table as well.

Note that in the mapping from PETSCII to ZSCII, you must also convert any accented characters from uppercase to lowercase, i.e. for Swedish PETSCII Å is mapped to a ZSCII ä.

If you build a game using Ozmoos's Debug mode (Uncomment the 'DEBUG' line near the start of make.rb), Ozmoos will print the hexadecimal ZSCII codes for all characters which it can't print. Thus, to create mappings for a game in a new language, you can start by running it in Debug mode to see the ZSCII character codes in use.

Assembly Flags

The flags described here can be set on the Acme commandline using the syntax
`-D[flag]=1`

General flags

`BGCOL=n`
`BGCOLDM=n`

Set the background colour for normal mode and darkmode.

`BORDERCOL=n`
`BORDERCOLDM=n`

Set the border colour for normal mode and darkmode.

`CACHE_PAGES=n`

Set the minimum number of memory pages to use for the cache (used to buffer pages otherwise residing at `$D000-$FFFF`).

`CHECK_ERRORS`

Enable some extra checks for runtime errors, making the interpreter a little bigger.

`CONF_TRK=n`

Point out the track on disk where the interpreter must look for the configuration information.

CURSORCOL=**n**
CURSORCOLDM=**n**

Set the cursor colour for normal mode and darkmode.

COL2=**n**
COL3=**n**
COL4=**n**
COL5=**n**
COL6=**n**
COL7=**n**
COL8=**n**
COL9=**n**

Replace color in Z-machine palette with a certain colour from the C64 palette.

CONF_TRK=**n**

Tell the interpreter on which track the config blocks are located (in sector 0 and 1).

CURSORCHAR=**n**

Use this character code as the cursor.

CUSTOM_FONT

Tell the interpreter that a custom font will be used.

FGCOL=**n**
FGCOLDM=**n**

Set the foreground colour for normal mode and darkmode.

FREE_SAVE_BLOCKS=**n**

Set the number of blocks the interpreter should expect to be available on a blank save disk. This number is used to calculate the number of save slots that can fit.

LURKING_HORROR

If the game uses sound, enable the code that defines assumptions we need to make for the Lurking Horror, i.e. use a hardcoded table defining which sounds have repetition enabled, as the Z-code opcode to play sound doesn't support setting this parameter in z3.

MAJOR_VERSION_NO

The major version number of the interpreter.

MINOR_VERSION_NO

The minor version number of the interpreter.

NOSCROLLBACK

Don't include the scrollbar buffer feature.

NOSECTORPRELOAD

Don't load any parts of static memory into RAM from disk sectors on boot. If make.rb can decide that this won't be needed anyway, it will set this flag, to make the interpreter a little smaller.

OPTIMIZE_VMEM

Enable continuous virtual memory optimization, moving vmem blocks around to make sure the most used blocks are in unbanked memory. This can only be used on C64 and C128, and has no effect when the REU Boost virtual memory model is activated, i.e. when the interpreter has been built with the REUBOOST flag and the player chooses to use an REU for caching game data.

PREOPT

Build the interpreter in PREOPT mode (used to pick which virtual memory blocks should be preloaded into memory when the game starts).

REUBOOST

Add support for an alternate virtual memory model which is simpler and faster, for use with REU. When the story file is stored in an REU, copying a page from REU to RAM is so fast that it doesn't pay off to spend a lot of time figuring out which vmem block to replace. Defining REUBOOST typically makes games played with an REU about 10% faster.

SCROLLBACK_RAM_PAGES=n

This value is typically between 24 and 48. Reserve this many pages at the top of RAM bank 0 for a scrollback buffer. However, if an REU is detected and used for scrollback at runtime, this RAM buffer should not be allocated. If the interpreter is built with REUBOOST, and there is room for the story file in the REU, and the user chooses to load the story file into the REU, this won't matter, because the top 12 KB of RAM bank 0 are unused in this scenario anyway. SCROLLBACK_RAM_PAGES is not used or supported for MEGA65, since it always uses AtticRAM for scrollback buffer.

SLOW

Prioritise small size over speed, making the interpreter smaller and slower. For X16, C128 and Plus/4, this is enabled by default and can't be disabled, due to their memory models.

SOUND (MEGA65 only)

Include support for sound (.wav sample files playback).

SPLASHWAIT=n

Set how many seconds the splash screen should show. If this is set to 0, disable the splash screen completely, making the binary slightly smaller. On the C64, the splash screen resides in the virtual memory buffer, which means story_start and the maximum amount of dynamic memory aren't affected for a virtual memory build, but the empty space does allow Exomizer to compress the boot file better.

STACK_PAGES=n

Set the number of memory pages to use for stack. Can't be less than 2 for a mode P build, or less than 4 for any other build, since the stack holds initialization code on boot. 4 is good for pretty much any ZIL or Inform 6 game. Inform 7 games expect a 64 page stack, and are likely to crash at some point if they get a stack as small as 4 pages.

STATCOL=n

Set the statusline colour. (only for z1-z3).

TARGET_C64
TARGET_X16
TARGET_C128
TARGET_PLUS4
TARGET_MEGA65

Set the target platform.

TERPNO=*n*

Set the interpreter number (0-19) as reported by the interpreter to the game. Default is 2 (Apple II) for Beyond Zork and 8 (C64) for other games.

TRACE

Allocate a page of memory to keep a record of the last 64 instructions that were executed. If both TRACE and DEBUG are defined, and the game crashes with a fatal error, the game prints out the ten last instructions that were executed (Z-machine address and opcode).

UNDO

Include support for Undo. Unless UNDO_RAM is also defined, this uses REU on C64 and C128, video RAM on Commander X16, and Attic RAM on MEGA65. Undo is not available on Commodore Plus/4.

UNDO_RAM

Include support for Undo in RAM (only on C128).

USE_BLINKING_CURSOR=*n*

Make the cursor blink during player input, where *n* is the number of jiffies between blinks.

USE_HISTORY=*n*

Enable command history, and allocate at least *n* bytes for storing it. Since the code to support command history is small, and since the storage is placed between the interpreter and the next thing in RAM (virtual memory buffers or stack or virtual memory) which needs to be page-aligned anyway, this can sometimes be enabled without causing `story_start` to move up.

VMEM

Utilize virtual memory. Without this, the complete game must fit in C64 RAM available above the interpreter, all in all about 50 KB. Also check section “Virtual memory flags” below.

X_FOR_EXAMINE

Enable support for automatically changing the word “X” to “EXAMINE”, when it’s typed as a verb by the player.

Z1
Z2
Z3
Z4
Z5
Z6
Z7
Z8

Build the interpreter to run Z-machine version 1, 2, 3, 4, 5, 6, 7 or 8.

Z6_ECM_MODE

C64 and version 6 only. Use the VIC-II’s Extended Color Mode, so that each of the eight windows can have its own background colour. Set by make.rb’s -ecm switch. See “Version 6”.

Z6_FCM_MODE

MEGA65 and version 6 only. Use the VIC-IV’s Full Colour Mode, with 16-bit character codes, giving a 320x200 screen of 40x25 cells. Set by make.rb’s -fcm switch. See “Version 6”.

Z6_PICTURES

MEGA65 and version 6 only, and requires Z6_FCM_MODE. The game’s pictures are on the disk and are drawn. Set by make.rb’s -pics switch. See “Version 6”.

DANISH_CHARS
FRENCH_CHARS
GERMAN_CHARS
ITALIAN_CHARS
SPANISH_CHARS
SWEDISH_CHARS

Map national characters in ZSCII to their PETSCII equivalents. No more than one of these can be enabled.

Internal Flags

These flags are set and used internally in Ozmoo, depending on the settings of the general flags, especially the TARGET_??? flag.

COMPLEX_MEMORY

If COMPLEX_MEMORY is set, then the Z-machine dynamic memory is not directly accessible in RAM, so we can't use simple lda/sta to access it. Instead, special routines are required. For C128 and X16, this involves selecting the right RAM bank, for MEGA65 it involves far memory access, while for Plus/4 it just means we need to bank out ROM which normally covers large areas of RAM during execution. Plus/4 has COMPLEX_MEMORY set, but not FAR_DYNMEM.

FAR_DYNMEM

If FAR_DYNMEM is set, then the Z-machine dynamic memory is not in the part of RAM which can be directly accessed with 16-bit addresses. This means we can't use simple lda/sta operations, and we can't assume that dynamic memory starts at the story_start marker. Thus, additional code is needed to read and write to dynamic memory. A platform that has FAR_DYNMEM set must also have COMPLEX_MEMORY set.

HAS_SID

If the target has a SID chip for sound generation.

SUPPORT_80COL

If the target has an 80 column display.

SUPPORT_REU

If the target potentially supports extended memory of some sort, to be used as a RAM disk for faster game play.

VMEM_END_PAGE=n

This is the page where virtual memory storage ends, i.e. the last page that can be used by the virtual memory system plus one. For the C64, the last page is \$FF (address FF00–FFFF), so VMEM_END_PAGE is \$00.

Debug flags

(If any of the flags in this section are enabled, DEBUG is automatically enabled too.)

DEBUG

Print debug information, print descriptive error messages, store a trace of the instructions which have been executed and print them in case of an exception. Also check section “Debug flags” below.

BENCHMARK

When the game starts, replay a number of colon-separated commands which have been stored with the interpreter (in the file text.asm). Charcode 255 in this command sequence means print the number of jiffies since the computer was started.

COUNT_SWAPS

Keep track of how many vmem block reads have been done.

PRINT_SWAPS

Print information whenever a memory block is loaded into memory.

TRACE_FLOPPY

Print trace information for floppy operations.

TRACE_FLOPPY_VERBOSE

Print verbose trace information for floppy operations.

TRACE_PRINT_ARRAYS

Print the contents of the input array and the parse array whenever a read command has been performed.

TRACE_READTEXT

Print trace information for opcode read/aread/sread.

TRACE_SHOW_DICT_ENTRIES

Print which dictionary words are checked when trying to match a word, and if the input word was less than or greater than that word.

TRACE_TOKENISE

Print trace information for the tokenization process.

TRACE_VM

Print trace information for the virtual memory system.

VICE_TRACE

Send the last instructions executed to Vice, to aid in debugging.

VIEW_STACK_RECORDS

Print whenever the stack size hits a new high, or when the number of bytes pushed onto the stack within a frame reaches a new high.

Configuration blocks

Interpreter configuration is stored in two blocks on the boot disk (track 1, sector 0-1). The interpreter loads this information when it starts.

Contents:

4 bytes: Game signature: A 32-bit random number.

--- Disk information block ---

1 byte: Number of bytes used for disk information, including this byte

1 byte: Interleave (in range 1-21)

1 byte: Number of save slots which can fit on a disk (in range 2-132)

1 byte: Number of disks used (disk 0 is the save disk(s),
disk 1 .. are game disks

For each disk:

1 byte: n: Number of bytes used for this entry, including this byte.

1 byte: Device#. Should be 8,9,10,11 or 0, meaning it's not decided
yet - it's up to the terp to set the device#.

2 bytes: Last story block # + 1 (high endian) on this disk

1 byte: t: Number of tracks used for story data

(If this number is > 0, this is a story disk!)

t bytes: Sectors used for story data:

Bit 0-4: # of sectors used.

Bit 5-7: First sector# used.

(Example: %01010000: Use 16 sectors, starting with #2)

x bytes: Disk name in PetSCII. Special codes:

0: End of string

128: "Boot "

129: "Story "

130: "Save "

131: "disk "

--- Vmem information block ---

2 bytes: Number of bytes used for vmem information, including this byte (high endian)

1 byte: Number of vmem blocks suggested for initial caching

1 byte: Number of vmem blocks already preloaded
x bytes: vmap_z_h contents for the suggested blocks
x bytes: vmap_z_l contents for the suggested blocks

Typical size for a 3-disk (+ save disk) game:

$4 + 1 + 1 + 1 + 1 + 2 * (1 + 1 + 2 + 1 + 40 + 5) + 2 * (1 + 1 + 2 + 1 + 0 + 3) + 1 + 1 + 1 + 100 + 100 = 8 + 2 * 50 + 2 * 8 + 203 = 8 + 100 + 16 + 203 = 327$ bytes

An interpreter needs to reserve this space for disk information:

z1/z2/z3: $1 + 1 + 1 + 1 + 2 * (1 + 1 + 2 + 1 + 0 + 3) + (1 + 1 + 2 + 1 + 40 + 6) = 4 + 2 * 8 + 51 = 71$ bytes

z4/z5/z8: $1 + 1 + 1 + 1 + 2 * (1 + 1 + 2 + 1 + 0 + 3) + 2 * (1 + 1 + 2 + 1 + 40 + 5) = 4 + 2 * 8 + 2 * 50 = 120$ bytes (* see Double 1571 drive support below)

1571 drive support:

- z1/z2/z3 games: No change
- z4/z5 games: A single disk using only track 1-53 can hold all story data. Disk information will then fit in less than the number of bytes stated above.
- z8 games: Disk information for a single drive game will fit in the number of bytes stated above.

Double 1571 drive support:

- z1/z2/z3/z4/z5 games: Any game fits in less than the amount of bytes calculated for a 3-disk game above.
- z7/z8 games: We need an additional 28 bytes to fit a full-size game. Thus, we allocate 150 bytes, to have some margin.

1581 drive support: This system should work, without extending the limits above, using only 31 sectors per track for story data, while allowing z8 games of up to 512 KB in size on a single disk. If we want to store several games on a single 1581 disk, we should add a track offset in each disk entry (i.e. saying that tracks below track x are considered empty).

Printer Support

Currently Ozmoos doesn't have any support for printers, but hooks exist that could be used to add such functionality.

Infocom provided printer output through the transcript command, which redirected text output to output stream 2. `stream.asm` implements streaming, and `stream__print__output` can be extended to handle stream 2 and send the bytes to the printer.

We currently have no hardware to test printer support on, and no immediate plans to provide such support. However, we are happy to accept patches.

Runtime Errors

These are the runtime errors that may occur when running Ozmoos:

- `ERROR_UNSUPPORTED_STREAM = 1`

The Z-machine supports certain streams for input and output of text. If a program tries to open a stream number which is not defined or which is not supported by Ozmoos, this error occurs.

- `ERROR_CONFIG = 2`

The boot disk of a program built with Ozmoos has config information on sector 1:0 and 1:1 (unless it's built as a single file program). This error means there seems to be something wrong with this information. The information is copied to memory when the game boots and then used whenever a block of game data needs to be copied from disk, so the problems may be discovered when the boot disk is no longer in the drive.

- `ERROR_STREAM_NESTING_ERROR = 3`

The Z-machine has a stack for memory streams. If the program tries to pull items from this stack when it is empty, this is the resulting error.

- `ERROR_FLOPPY_READ_ERROR = 4`

There was a problem reading from the disk.

- `ERROR_STACK_FULL = 6`

The program tried to make a routine call or push data onto the stack when there was not enough room. This means there is a bug in the program, or it needs a bigger stack. Stack size can be set with a commandline parameter to `make.rb`.

- `ERROR_STACK_EMPTY = 7`

The program tried to pull a value from the stack when it was empty.

- `ERROR_OPCODE_NOT_IMPLEMENTED = 8`

An unknown Z-machine opcode was encountered. This can happen if the wrong disk is in the drive when Ozmo tries to retrieve a block of program code.

- `ERROR_USED_NONEXISTENT_LOCAL_VAR = 9`

Each routine in Z-code has between 0 and 15 local variables. If an instruction references a local variable number which is not present in this routine, this is what happens. Normally, a compiler like Inform doesn't let the programmer write code which can cause this, unless you skip the highlevel language and write Z-machine assembler.

- `ERROR_BAD_PROPERTY_LENGTH = 10`

The program tried to use an object property of an illegal length, where a property value has to be one or two bytes.

- `ERROR_UNSUPPORTED_STORY_VERSION = 11`

The first byte of the story file must match the Z-machine version for which the interpreter was built (3, 4, 5 or 8). Otherwise, this occurs.

- `ERROR_OUT_OF_MEMORY = 12`

The program referenced memory which is higher than the last address in the story file.

- `ERROR_WRITE_ABOVE_DYNMEM = 13`

The program tried to write to memory which is not part of dynamic (RAM) memory.

- `ERROR_TOO_MANY_TERMINATORS = 15`

A dictionary in a Z-machine program holds a list of terminating characters, which are used to separate words. It is illegal for this list to hold more than ten characters. If it does, this error occurs.

- `ERROR_NO_VMEM_INDEX = 16`

The `vmem_oldest_index` is only populated inside the loop if we find a non-PC vmem block older than the initial `vmem_oldest_age` of \$ff. That should always happen, but to assert that the code is correct the debug version of Ozmoos checks that `vmem_oldest_index` is valid and issues `ERROR_NO_VMEM_INDEX` if not.

- `ERROR_DIVISION_BY_ZERO = 17`

An attempt was made to divide a number by zero.